



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT213

Mid Term Examination

Date: 6/21/2025

Course Title: Statistics and Probability

Semester: Fall 2025

Total Marks: 30

Total Time: 1:30 Hours

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- Q1.** (a) Define discrete and continuous probability distribution. CLOs Marks
CLO1 [2]
(b) The following data show the suicides of 1096 women in 8 cities in a country during 14 years. CLO3 [4]

No. of suicides	0	1	2	3	4	5	6	7
Frequency	364	376	218	89	33	13	2	1

Fit a poison distribution to the above data.

- (c) Six unbiased coins are tossed simultaneously. What is the probability of getting CLO3 [4]
i. exactly three heads?
ii. at least four heads?

- Q2.** (a) What is a sample space? CLO1 [1]
(b) Two unbiased dice are tossed simultaneously. What is the probability of getting a total of point 9 or odd numbers from the both dice. CLO2 [4]
(c) Find the mean deviation from the mean and mean deviation from the median from the frequency distribution CLO3 [5]

Class interval of working hours	Number of employees (f_i)
30-35	8
35-40	3
40-45	14
45-50	22
50-55	7
55-60	6
Total	60

- Q3.** (a) State and prove Bayes theorem CLO2 [4]
(b) A bucket contains 6 blue, 4 green and 5 red balls. Sudheer is asked to pick 2 balls randomly from the bucket without replacement and then one more ball is to be picked. What is the probability he picked 2 green balls and 1 blue ball? CLO3 [3]
There are two identical boxes containing respectively 6 white and 4 red balls, 4 white and 6 red balls. A box is chosen at random and a ball is drawn at random [3]
(c) from it. If the ball is red, what is the probability that it is from the second box? CLO3

- Q4.** (a) Find the relation between arithmetic mean, geometric mean and harmonic mean. CLO2 [2]



HAMDARD UNIVERSITY BANGLADESH

- (b) Find the mean deviation from the mean and standard deviation of the observations like $a, a + d, a + 2d, \dots, a + 2nd$. Prove that, the latter is greater than the former. CLO3 [4]

- (c) Find the 35th percentiles and 7th deciles from the frequency distribution table given below CLO3 [4]

Class Interval of Working Hours	Number of Employees (f_i)
30-35	3
35-40	4
40-45	12
45-50	20
50-55	5
55-60	6
Total	40

Q5.

- (a) Define standard deviation. [1]
(b) Using the frequency distribution given in the following table, find out the quartile deviations [5]

Frequency distribution table:

Class Interval (Variable)	Frequency (f_i)
0-5	4
5-10	6
10-15	11
15-20	12
20-25	20
25-30	9
30-35	8
Total	70

- (c) A student while calculating mean and standard deviation of 25 observations obtained mean 56 and standard deviation 2. At the time of checking, it was found that he has copied 64 instead of 46. What would be actual value of mean and standard deviation? CLO2 [4]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE215: Data Structures

Batch: 26th

Semester: Spring 2025

Date: 30/06/2025

Mid Term Examination

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following:

5x6= 30

- 1 (a) Outline the differences between primitive and non-primitive data structure. CLO 1 3
- (b) Explain the basic operations of data structures with example. CLO 1 3
- 2 (a) Given Array $A = \{29, 18, 4, 3, 12, 7, 11, 8, 25\}$, apply the Merge sort algorithm to sort the elements in ascending order. CLO 2 3
- (b) Summarize the time complexities of Insertion Sort, Selection Sort, and Bubble Sort. CLO 2 3
- 3 (a) Change the following infix expression to prefix and postfix expression: $(A+B)*C-D/E$. CLO 3 3
- (b) Convert the following expression to postfix using Stack: $A+B*(C^D-E)/F+G$. CLO 4 3
- 4 (a) Use Radix Sort Algorithm to sort the elements in the array $A = \{321, 542, 783, 874, 65, 987, 123\}$. Explain the sorting process. CLO 2 3
- (b) Apply selection sort algorithm to the array $A = \{56, 44, 92, 18, 71, 33, 51, 69\}$, ordering the numbers from smallest to largest. CLO 2 3
- 5 (a) Apply Binary Search Algorithm to locate the element 31 in the sorted array $[5, 10, 15, 20, 25, 30, 35, 40]$. Show the process step by step. CLO 2 3
- (b) Perform Bucket Sort Algorithm on the following list: 29, 34, 47, 15, 7, 88, 92, 61, 5, 75 and show the intermediate steps, assuming each bucket can hold a range of 10 values. CLO 2 3
- 6 (a) Describe the algorithmic approach of ternary search. CLO 2 3
- (b) List the key differences between binary search and ternary search. CLO 2 3
- 7 (a) Demonstrate the formal algorithm for traversing a singly linked list. CLO 1 2
- (b) Given a singly linked list with the elements $\{4, 8, 12, 16\}$, insert a new node with the value 10 at the 3rd position in the list. CLO 1 2
- (c) Given a singly linked list with the elements $\{10, 20, 30, 40, 50\}$, delete the node at the 3rd position and return the updated list. CLO 1 2

Hamdard University Bangladesh
B.Sc. Mid-term Examination Spring-2025
Department of Computer Science and Engineering
Course – CSE213 (Digital Electronics)

Full Marks: 30

Time: 1.30 H

(Answer any three questions)

		CLOs	Marks
1	a) What is binary number and radix? Write down the impotency of binary number.	CLO1	3
	b) You have 30 tokens, how many numbers can be represented in Binary, Ternary and Decimal numbering system?	CLO2	3
	c) Explain gray code. Convert $(0111)_2$ into gray code and $(0101)_8$ into binary code.	CLO1	4
2	a) Define and explain Moore's laws.	CLO1	3
	b) Draw the circuit diagram of Resistor-Transistor Logic (RTL), Diode-Transistor Logic (DTL), and Bipolar Transistor-Transistor Logic (TTL).	CLO3	7
3	a) What do you understand by MOS? Construct the p-channel MOSFET.	CLO1	4
	b) Explain about CMOS.	CLO1	3
	c) Draw a clock signal. Write down the application of clock signal.	CLO2	3
4	a) What is Multivibrator? Write down some application of Multivibrator.	CLO1	2
	b) Discuss about different types Multivibrators.	CLO2	3
	c) Draw and describe the working principle of astable multivibrator.	CLO3	5
5	a) What do you understand by sequential circuit? What is memory element?	CLO1	2+1
	b) Draw and describe the operation of S-R latch with truth table.	CLO2	3
	c) Why Master Slave Flip Flop? Explain the working principle of a Master Slave flip-flop.	CLO3	4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science , Engineering and Technology Bachelor of Science in Computer Science and Engineering

BAN211: History of Emergence of Bangladesh

Mid-Term Examination

Spring 2025

26th Batch
18 June 2025

3rd Semester
Time: 1 hour 30 minutes

Total Marks: 30

Part I: Short Questions

1. What were the reasons behind the arrival of Europeans in Indian Subcontinent? 05
2. Explain how the fall of Nawab Siraj ud-Daulah changed the political and economic situation in Bengal. 05
- Or
3. Describe the main features of the Permanent Settlement and how it changed land ownership in India.
4. Explain how British development policies in India were used as a tool of control and dominance over Indian society. 05
- Or
5. Describe the military factors that contributed to the Sepoy Mutiny.
6. Was the Partition of Bengal justified by the British as an administrative necessity, or was it a strategic move to divide Indian unity? Evaluate with reasons. 05
- Or
7. How successful was the Khilafat and Non-Cooperation Movement in maintaining Hindu-Muslim unity, and what factors contributed to its eventual breakdown? Evaluate with examples.

Part II: Broad Questions

8. Analyze how the political and social experience of Muslims under Congress rule from 1937 to 1939 contributed to the growing demand for Pakistan. 10
- Or
9. Analyze the key objectives and challenges of the United Bengal Plan and explain why it ultimately failed to gain support.



HAMMARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT217

Mid-term Examination

Date: 6/18/2025

Course Title: Ordinary & Partial Differential Equation

Semester: Spring 2025

Total Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any three (03) out of the following questions:

3 × 10 = 30

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- Q1. (a) Define homogeneous and non-homogeneous differential equations. Identify the which of the following is homogenous and which one is non-homogenous
- I. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = e^{-2x} \sec^2 x$
- II. $\frac{d^2y}{dx^2} + 7\frac{dy}{dx} = -10y$
- III. $\frac{dy}{dx} = \frac{y}{x} + x \sin \frac{y}{x}$
- (b) Solve the initial value problem $(x^2 + 3y^2)dx - 2xydy = 0$ when the initial condition is $y(2) = 1$
- (c) Solve the differential equation $4xydx + (x^2 + 1)dy = 0$
- (a) Using Bernoulli equation, solve the following differential equation
- i. $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$
- ii. $x^2dy + y(x+y)dx = 0$
- (b) Using variable separation method, solve $x^4\frac{dy}{dx} + x^3y = -\sec xy$
3. (a) Prove that the equation $[1 + \log(xy)]dx + \left(1 + \frac{x}{y}\right)dy = 0$ is an exact differential equation.
- (b) Check whether this differential equation $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$ is exact or not. Also solve the equation.
- (c) Use an integrating factor to solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$
- Q4 (a) Define degree and order of an ordinary differential equation. Also find the degree and order of the following differential equations
- i. $x^2\left(\frac{d^2y}{dx^2}\right)^3 + y\left(\frac{dy}{dx}\right)^4 + y^4 = 0$
- ii. $\left(\frac{d^3y}{dx^3}\right)^3 + \left(\frac{dy}{dx}\right)^4 + x^3 = 0$
- (b) Solve the following non-homogeneous second-order differential equations:
- i. $(D^2 + 6D + 9)y = \frac{e^{-3x}}{x^3}$
- ii. $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = e^{2x} \sin x$
- Q5 (a) Find the general solution of the following differential equation:



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- i. $(D^3 - 3D^2 + 4D - 2)y = 0$
- ii. $(D^2 + 4D + 5)y = 0$
- iii. $(D^2 - 8D + 16)y = 0$

(b) Solve: $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 6e^{2x} + 7e^{-3x} - \log 3$

CLO3

[4]



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Engineering & Technology
Program CSE

ACT 215: Financial and Managerial Accounting
Mid-Term Examination
Spring 2025

23.06.2025

Time: 1 Hour 30 Minutes

Total Marks: 30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.
- Answer Any Three (03) from the following set (No. 1-5).

No.	Question/ Statement	CLOs	Marks
1.	Mark Miller started his own delivery service, Miller Deliveries, on June 1, 2024. The following transactions occurred during the month of June.	2	
June 1	Mark invested \$17,000 cash in the business.		
2	Purchased a used van for deliveries for \$12,000. Mark paid \$2,000 cash and signed a note payable for the remaining balance.		
3	Paid \$500 for office rent for the month.		
5	Performed \$4,450 of services on account.		
9	Withdrew \$200 cash for personal use.		
12	Purchased supplies for \$150 on account.		
15	Received a cash payment of \$1,650 for services provided on June 5.		
17	Purchased gasoline for \$100 on account.		
20	Received a cash payment of \$1,500 for services provided.		
23	Made a cash payment of \$800 on the note payable.		
26	Paid \$250 for utilities.		
29	Paid for the gasoline purchased on account on June 17.		
30	Paid \$1,050 for employee salaries.		

Instructions:

- Show the effect of above-mentioned transactions on the accounting equation using the following heads:
Cash, account receivable, supplies, delivery van, notes payable, accounts payable, Miller capital.
- Construct income statement and balance sheet of June 1, 2024.

7

2. Hyzer Disc Golf Course was opened on March 1 by Barry Schultz. The following selected events and transactions occurred during March:

Mar. 1	Invested \$20,000 cash in the business.	
3	Purchased Heeren's Golf Land for \$16,000 cash. The price consists of land \$12,000, shed \$2,000, and equipment \$1,000. (Make one compound entry.)	
5	Advertised the opening of the driving range and miniature golf course, paying advertising expenses of \$500.	
6	Paid cash \$600 for a one-year insurance policy.	
10	Purchased golf discs and other equipment for \$1,050 from Innova Company payable in 30 days.	
18	Received \$390 in cash for golf fees earned.	
19	Sold 100 coupon books for \$12 each. Each book contains 4 coupons that enable the holder to play one round of disc golf.	
25	Withdrew \$800 cash for personal use.	
30	Paid salaries of \$250.	
30	Paid Innova Company in full.	
31	Received \$200 cash for fees earned.	

Journalize the March transactions.

- | | | |
|----|--|-----|
| 3. | a) Define accounting as per AICPA. | 10 |
| | b) Discuss the functions of accounting. | 1 2 |
| | c) Explain the different users of accounting information | 4 |

4. Barone's Repair Shop was started on May 1 by Nancy Barone. A summary of May transactions is presented below.

May 1	Invested \$10,000 cash to start the repair shop.	
4	Purchased equipment for \$9,500 cash.	
7	Paid \$400 cash for May office rent.	
9	Paid \$500 cash for supplies.	
10	Incurred \$280 of advertising costs in the Beacon News on account.	
16	Received \$5,100 in cash from customers for repair service.	
18	Withdrew \$1,200 cash for personal use.	
23	Paid part-time employee salaries \$2,500.	
27	Paid utility bills \$140.	
29	Provided repair service on account to customers \$750.	
31	Collected cash of \$220 for services billed in transaction (10).	

Journalize the March transactions.

5. a) Define management accounting.
Describe the nature of management accounting.
- b) Compare between management accounting and financial accounting.